

Choosing the random variable

A manager asks whether a batch of 80 independent jobs averages more than 7 minutes per job.

Should the probability model describe one duration, a mean, or a sum?

One observation

Model: independent task durations follow an exponential distribution with mean 6 minutes.

What is the probability that one task takes longer than 8 minutes?

An average of observations

Model: independent task durations are exponential with mean and SD 6 minutes. A batch contains 100 tasks.

What is the approximate probability that its mean exceeds 7 minutes?

Comparing the two events

Model: independent task durations are exponential with mean 6 minutes.

Would one task longer than 7 minutes be as unusual as 100 tasks averaging longer than 7 minutes?

A total duration

Model: independent task durations are exponential with mean and SD 6 minutes. A batch contains 100 tasks.

What is the approximate probability that its total exceeds 700 minutes?

An equivalent threshold

A batch contains exactly 100 tasks.

Why do a mean exceeding 7 minutes and a total exceeding 700 minutes describe the same event?

A mean percentile

Model: independent exponential task durations have mean and SD 6 minutes. Samples contain 100 tasks.

What is the approximate 90th percentile of sample mean duration?

A sum percentile

Model: independent exponential task durations have mean and SD 6 minutes. Samples contain 100 tasks.

What is the approximate 90th percentile of total duration?

A uniform population

Model: independent processing times follow $U(2,10)$, in seconds.

What population mean and SD are needed to analyze a sample mean?

A uniform mean

Model: independent processing times follow $U(2,10)$. Samples contain 64 jobs.

What approximate model describes their mean time?

A lower-tail mean

Model: independent processing times follow $U(2,10)$. Samples contain 64 jobs.

What is the approximate probability that their mean is below 5.5 seconds?

A uniform total

Model: 64 independent processing times follow the uniform distribution $U(2,10)$, in seconds.

What is the approximate probability that their total is below 350 seconds?

A uniform mean percentile

Model: 64 independent times follow $U(2,10)$, in seconds.

What is the approximate 90th percentile of their sample mean?

A uniform total percentile

Model: 64 independent times follow $U(2,10)$, in seconds.

What is the approximate 90th percentile of their total?

Quartiles and IQR of sample means

Model: independent observations have mean 20 min and SD 4 min. Samples contain 100 observations. Use a normal approximation.

What are the sample-mean quartiles and IQR?

Quartiles and IQR of sums

Model: independent observations have mean 20 min and SD 4 min. Samples contain 100 observations. Use a normal approximation.

What are the sum quartiles and IQR?

A central interval for totals

Synthetic independent durations are exponential with mean 8 min. Sum 64 durations; use the normal approximation.

Between which two totals does the middle 80% lie?

Checking the approximation

Model: independent processing times follow $U(2,10)$. A normal approximation is used for a mean of 64 times.

Can the true sample mean ever be below 2 seconds?

A population shape problem

Synthetic data show strong right skew and several extreme outliers. A random sample contains 12 observations.

Is a normal approximation for its mean automatically justified?

A selection problem

A large sample includes only files that finished downloading successfully.

Can a small standard error remove bias caused by excluding failed downloads?

The law of large numbers

Under suitable conditions, the average of independent observations approaches the population mean as sample size grows.

Does this guarantee that every larger sample mean is closer to μ than the previous one?

Two different ideas

The law of large numbers concerns closeness to μ ; the central limit theorem concerns the distribution of standardized averages.

Which idea supports approximating sample-mean tail probabilities with a normal curve?

A repeated-sample experiment

Use synthetic independent draws from $U(0,12)$, in minutes. For each n , collect many sample means.

What changes would you expect in histograms for $n=1$, $n=5$, and $n=50$?

Normal approximation to the binomial

A binomial distribution counts successes in n independent constant- p trials. For sufficiently large expected counts, a normal distribution can approximate it.

What mean and standard deviation should approximate $B(100, 0.4)$?

Approximation conditions

OpenStax requires $np > 5$ and $nq > 5$, with $q = 1 - p$; both at least 10 generally improve the approximation.

Would $B(100, 0.02)$ meet these conditions?

The approximating normal model

For $X \sim B(n, p)$, use a continuous Y with the same mean and standard deviation.

$$Y \sim N(np, \sqrt{np(1 - p)})$$

For $X \sim B(100, 0.4)$, what are the parameters of Y ?

Continuity correction factor

The continuity correction factor 0.5 matches each discrete integer k to a continuous interval from $k - 0.5$ to $k + 0.5$.

Why should a normal approximation to $P(X = 40)$ use an interval rather than one point?

At least: continuity correction

Model: $X \sim B(100, 0.4)$. Approximate using $Y \sim N(40, \sqrt{24})$.

Which boundary and normal tail approximate $P(X \geq 45)$, and what probability results?

At most: continuity correction

Model: $X \sim B(100, 0.4)$. Approximate using $Y \sim N(40, \sqrt{24})$.

Which boundary and normal tail approximate $P(X \leq 45)$, and what probability results?

More than: continuity correction

Model: $X \sim B(100, 0.4)$. Approximate using $Y \sim N(40, \sqrt{24})$.

Which boundary and normal tail approximate $P(X > 45)$, and what probability results?

Fewer than: continuity correction

Model: $X \sim B(100, 0.4)$. Approximate using $Y \sim N(40, \sqrt{24})$.

Which boundary and normal tail approximate $P(X < 45)$, and what probability results?

Exactly: continuity correction

Model: $X \sim B(100, 0.4)$. Approximate using $Y \sim N(40, \sqrt{24})$.

Which two boundaries approximate $P(X = 45)$, and what probability results?

Checking the approximation

For $X \sim B(100, 0.4)$, use software to find the exact $P(X \geq 45)$.

How does it compare with the continuity-corrected normal approximation?

Predicting precision

Model: independent observations have population SD 12.

What sample size gives a standard error of 1?

An observed mean

Model: independent normal durations have mean 20 and SD 8 minutes. A synthetic sample of 64 has mean 22.

How many standard errors above the modeled mean is this result?

Unknown population mean

A synthetic random sample has $n=64$, mean 22 minutes, and sample SD 8 minutes.

Which population quantity would a confidence interval try to estimate?

A claim about the mean

A process is claimed to average 20 minutes. A random sample averages 22 minutes.

Why should sample size and variability matter before judging that claim?